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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Com (BM) DEGREE EXAMINATION APRIL 2021

ASCM301Q - BUSINESS STATISTICS

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2=20)

Answer ALL the questions

1. Define secondary data.

2. Calculate Median and Geometric Mean for the following data: 3, 6, 24, 48.

3. Define Arithmetic Mean.

4. Find the standard deviation of the set of numbers 3, 8, 6, 10, 12, 9, 11, 10, 12, 7.

5. Define correlation and give its limit.

6. Give regression equations.

7. State the uses of index numbers.

8. State the methods of constructing index numbers

9. State the uses of time series.

10. What are the components of time series?

SECTION - A (5 x 5 =25)

Answer **ALL** the questions

11. (a) Find Harmonic Mean for the following data.

X :	10	20	30	40	50
Y :	2	8	10	7	3

(Or)

(b) Find the Standard Deviation for the following data.

Number of days absent	0-10	10-20	20-30	30-40	40-50	50-60
Number of students	3	7	20	10	6	12

12. (a) Calculate Karl Pearson's Coefficient of Skewness for the given data.

X : 10 15 20 25 30

Y : 7 12 40 20 10

(Or)

(b) Calculate the quartile deviation from the following frequency distribution..

Value of the item	12	13	14	25	26	27	38	40
Frequency	2	3	5	8	7	3	2	1

Identify the better player and the more consistent player.

13. (a) Calculate coefficient of correlation by the method of concurrent deviation from the following data:

X	84	85	62	48	84	95	103	100	85	115
Y	20	23	19	21	25	25	28	27	26	30

(Or)

(b) Find the line of regression of y on x:

X	1	2	3	4	5	8	10
Y	9	8	10	12	14	16	15

14. (a) Construct the cost of living index number from the following group data:

Group	Group Index Number for a given year	Weights
Food	352	48
Clothing	200	10
House Rent	160	12
Fuel and Lighting	230	8
Miscellaneous	190	15

(Or)

(b) From the fixed base index numbers given below, obtain the chain base index numbers:

Year	Fixed base indices
2001	94
2002	98
2003	102
2004	95
2005	98
2006	100

15. (a) Compute seasonal indices from the following data.

Year / Quarter	1996	1997	1998	1999	2000
I	72	76	74	76	78
II	68	70	66	74	74
III	80	82	84	84	86
IV	70	74	80	78	82

(Or)

(b) State the uses of Time series.

SECTION - C (3 X 10 = 30)

Answer **any THREE** questions.

16. Find Mean, Median and mode for the following data.

Marks	: 0-10	10-20	20-30	30-40	40-50	50-60	60-70
No.of Students	: 4	12	28	20	10	6	2

17. Calculate the mean deviation about mean for the following data.

Class	0-10	10-20	20-30	30-40	40-5-0	50-60	60-70
Frequency	8	12	10	8	3	2	7

18. Calculate Spearman's Rank correlation for the following data.

X	300	450	800	250	500	975	475	400
Y	273	486	734	297	631	872	396	457

19. Using the following data, construct Fisher's ideal index and show how it satisfies Factor Reversal test and Time Reversal test.

Commodity	Price in rupees per unit		Number of units	
	Base year	Current year	Base year	Current year
A	8	12	6	4
B	10	12	8	8
C	14	18	4	4
D	4	2	6	10
E	10	14	10	8

20. Below are given the figures of production (in thousand tones) of a fertilizer factory.

Year	1996	1997	1998	1999	2000	2001	2002	2003
Production ('000 tonnes)	38	40	65	72	69	60	87	95

Fit a straight line by the least square method and tabulate the trend values.

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